Shavir Vallabh

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Final-year Computer Science student at the University of Pretoria with a passion for software engineering, cybersecurity, applied AI and high-performance computing.

Education

University of Pretoria
BSc Computer Science, Final Year Jan 2024 - Present

- Consistently maintained a position on the Dean's Merit List with a cumulative GPA of **80.55%.**
- Achieved distinctions in all computer science and core mathematics courses.
- Three-time recipient of the Fly@UP senior undergraduate award, in addition to a full scholarship and Golden Key recognition.
- Nominated as class representative for Introductory CS (COS 151), Discrete structures II (WTW 285) and Artificial Intelligence (COS 314).

Technical Skills & Interests

Languages: C/C++, C#, Python, PHP, Java, JavaScript/TS, Shell Scripting
Frameworks & General: Django, Node.js, React, Angular, MySQL, PostgreSQL
DevOps & Sysadmin: GitHub Actions, Docker, Kubernetes, Linux (Ubuntu, RHEL, Arch), Cluster Computing
Interests: Speedcubing, Reading, Console Modding & Homebrew, Electric Guitar, Gym

Experience

University of Pretoria (Department of Computer Science)
Software Developer & Sysadmin (TechTeam) Jan 2026 - Present

- Deployed a 5-node GPU cluster with 99.9% uptime for Big Data Science research, increasing compute power by over 400% over individual nodes, using Slurm, LDAP auth, and Grafana monitoring.
- Developed an internal rubric marking tool used for over 1 200 students with Django, enabling the creation of rubrics, as well as secure assessment of submissions.
- Reduced vulnerabilities on core university servers by 40%, through OS upgrades, firewall configuration, intrusion prevention measures, and server monitoring.

University of Pretoria
Teaching Assistant Jan 2025 - Dec 2025

- Served as a TA for *Introductory CS (*COS 151) and *Discrete structures II* (WTW 285).
- Responsible for marking homework assignments and test scripts; as well as presenting tutorials to classes of 100+ students.

University of Pretoria (IT Services)
Computer Lab Technician Jan 2024 - Dec 2025

- Provided technical support and troubleshooting for students and faculty, including software/hardware maintenance and setup assistance during examinations and academic events.

Achievements & Activities

NICIS Centre of High-Performance Computing
Team Captain, Student Cluster Competition 2025/26 Jul 2025 - Present

- Led a team in the 2025 and 2026 SCC selection round, competing against teams from 18 tertiary institutions, successfully earning a position in nationals both times.
- Built a virtual 3-node cluster on the cloud, compiled and tested a suite of different synthetic and scientific workloads.
- Designed a real cluster within a R650 000 budget and utilized this hardware in nationals.

South African National Research Network
Team Lead, Cyber Security Challenge 2026 Sept 2026

- Competed against 850+ students in various 'capture the flag' style challenges focused on digital forensics, penetration testing, reverse engineering, cryptography, and web exploitation.